CIV102 Project Team 502 Design Report

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Introduction

Using iterative processes, our team designed a beam bridge with a span of 1260mm, intended to support loading from a train, while utilising as little material as possible to maximise the strength to weight ratio. The design features a 140mm tall cross-section, with a double thickness top flange spanning 100mm, along with no bottom flange. The webs of the cross-section are 70mm apart, creating an overhang of 15mm on either side of the webs, with 11 diaphragms 150mm apart from each other. This design is predicted to hold a maximum load of 1053N, with an efficiency score of 0.1163.

Figure 1: Final design of the bridge

We went through 6 major iterations before landing on the final design, with each iteration being done to address the limiting failure modes of the previous iteration.

<u>Key Iterations</u>

- Iteration 1: Doubled the top flange thickness (1.27mm → 2,54mm) to address the failure mode of case 1 plate buckling on the top flange.
- Iteration 2: Removed the bottom flange to remove excess material, improving the strength to weight ratio, without harming the maximum load capacity of the final design.

- Iteration 3: The total height of the cross-section increased (77.54 mm → 100mm) to address the failure mode of case 1 plate buckling on the top flange, by increasing the second moment of inertia of the design.
- Iteration 4: Lowered the base width (80mm → 70mm) to further improve the factor of safety for case 1 plate buckling on the top flange by lowering the width of the constrained flange, this improves the efficiency, balances the factors of safety.
- Iteration 5: Increased the total height (100mm → 140mm), increasing the second moment of inertia to address the failure mode of compressive failure.
- Iteration 6: Decrease distance between diaphragms (400mm → 150mm) to address the failure mode of shear buckling.

Each iteration apart from iteration 6 was analysed using load case 1, consisting of 3 equally weighted train cars for a total of 400N, while only the final design to analysed using load case 2, to obtain the final maximum load capacity possible using the design.

The efficiency index used during the iterative process is a predictive measure of the strength to weight ratio of the design, as our design team used volume instead of weight to predict what the strength to weight ratio of each design would be in relation to each other, ad gives the design team an idea of how to effectively improve this.

Throughout the iteration process, the design team's philosophy prioritised addressing each limiting failure mode independently according to what each iteration requires, while maximising the second moment of inertia of the design, to optimise the strength to weight ratio of the final design.

Iteration (Design) 0:

```
%%% Define the Properties of the CrossSection %%%
thickness = 1.27;
basewidth = 80; %%% Must be Able to Hold The Trains
basethickness = thickness;
topflangethickness1 = thickness;
topflangethickness2 = 0;
totalbeamheight = 76.27;
topwidth = 100;
topwidthsecondlayer = 0;
tabwidth = 5;
%%% ---------------------------------------- %%%
%%%DEFINE THE DIAPHRAGMS
DiaphragmSpacing = 400;  %%% Max Diaphragm Spacing
NumberDiaphragms = 6;
%%% ---------------------------------------- %%%
```

Figure 2: An image showing the parameters used when running the code to evaluate design 0.

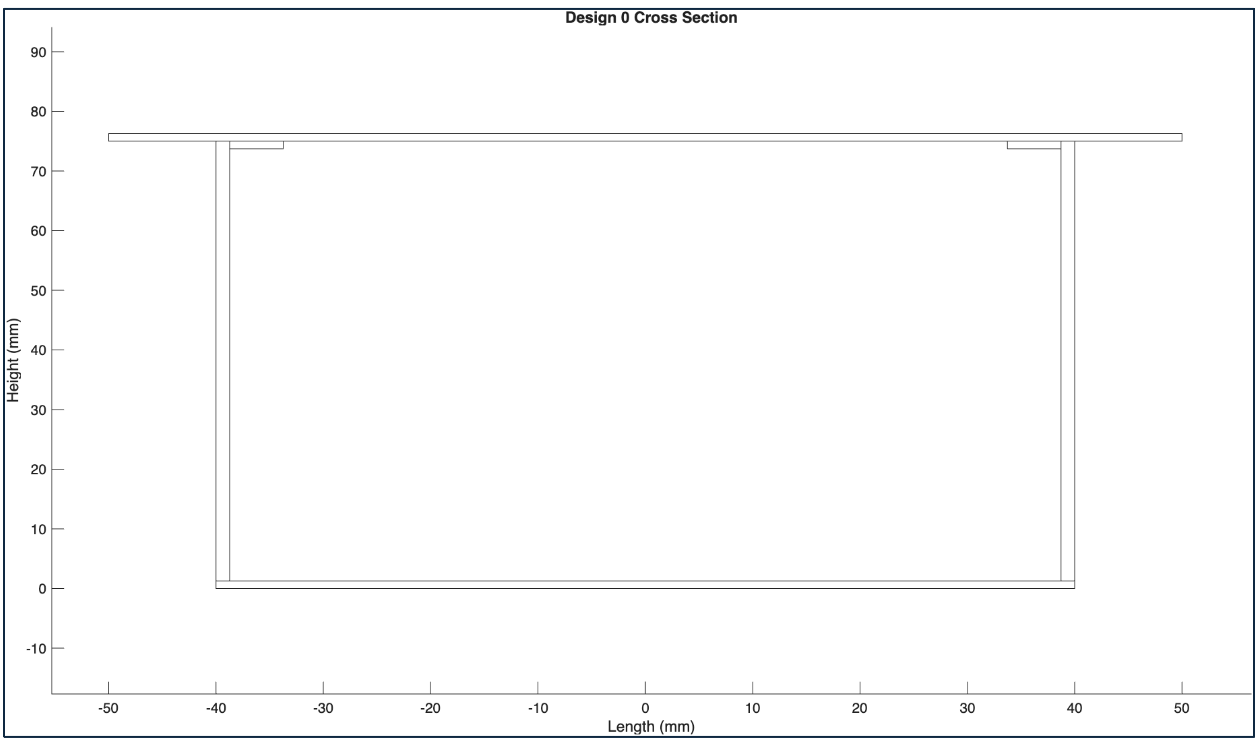

Figure 3: Image displaying the cross-section of Design 0.

Table 1: Table showing the important metrics of design 0

Maximum Load (N)	239
Percent of Material Used	55.6
Efficiency Index	0.0410
Centroidal Axis From Bottom (mm)	41.4
Moment of inertia ($\times 10^6 mm^4$)	0.418

Table 2: Table showing the factors of safety of design 0 with the failure mode highlighted

Compressive Failure of Top Flange	1.039
Tensile Failure of Top Flange	4.37
Shear Failure at Centroidal Axis	2.67
Shear Failure of Glue	7.49
Plate Buckling Case 1	0.598
Plate Buckling Case 2	4.07
Plate Buckling Case 3	5.29
Plate Buckling Case 4	3.40

Iteration 1: Double Top Thickness

- Double top piece thickness (1.27mm à 2.54mm) (stack pieces of matboard).

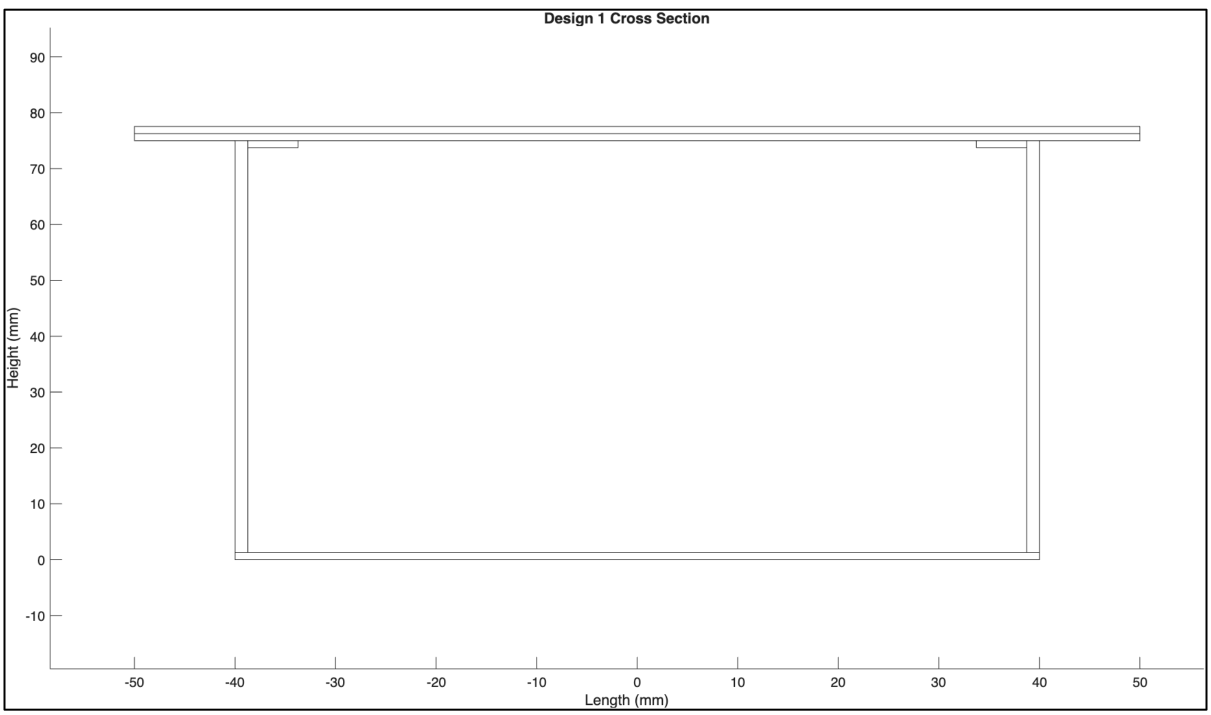

Figure 4: Image displaying the cross-section of iteration 1.

Table 3: Table showing the important metrics of iteration 1

Maximum Load (N)	669
Percent of Material Used	70.9
Efficiency Index	0.0901
Centroidal Axis From Bottom (mm)	49.5
Moment of inertia ($\times 10^6 mm^4$)	0.542

Table 4: Table showing all factors of safety for iteration 1 with the most likely failure mode highlighted.

Compressive Failure of Top Flange	1.674
Tensile Failure of Top Flange	4.73
Shear Failure at Centroidal Axis	2.70
Shear Failure of Glue	6.21
Plate Buckling Case 1	3.85
Plate Buckling Case 2	26.2
Plate Buckling Case 3	15.70
Plate Buckling Case 4	3.43

The thickness of the top piece was doubled, from 1.27 mm to 2.54mm, which would be done by stacking the matboard on top of each other. This addresses Design 0's failure mode of case 1 plate buckling occurring in the top flange. The buckling stress is directly proportional to the square of the thickness ($\sigma_{crit} \propto t^2$), additionally, it is inversely proportional to the second moment of inertia. Therefore, doubling said thickness increases the second moment of inertia improves the buckling resistance from a factor of safety of 0.598 to 3.85. This improved the maximum load capacity to 669N. This change also significantly raised the factor of safety against flexural tension, as second moment of inertia (I) increased to 5.42$\times 10^5 mm^4$, lowering the applied stress on the walls according to the equation $\sigma = \frac{My}{I}$, and therefore increasing said factor of safety from 1.039 to 4.73. The factor of safety of case 3 plate buckling of the walls as the buckling stress is proportional to the square of the thickness ($\sigma \propto t^2$), so doubling thickness dramatically increased said factor of safety from 5.29 to 15.70. For this iteration the compressive failure in the top flange is now the failure mode, despite the increase in the second moment of inertia.

Iteration 2: Remove the Base

- Remove bottom piece entirely.
- Reduces material used, no significant changes to lowest FOS.

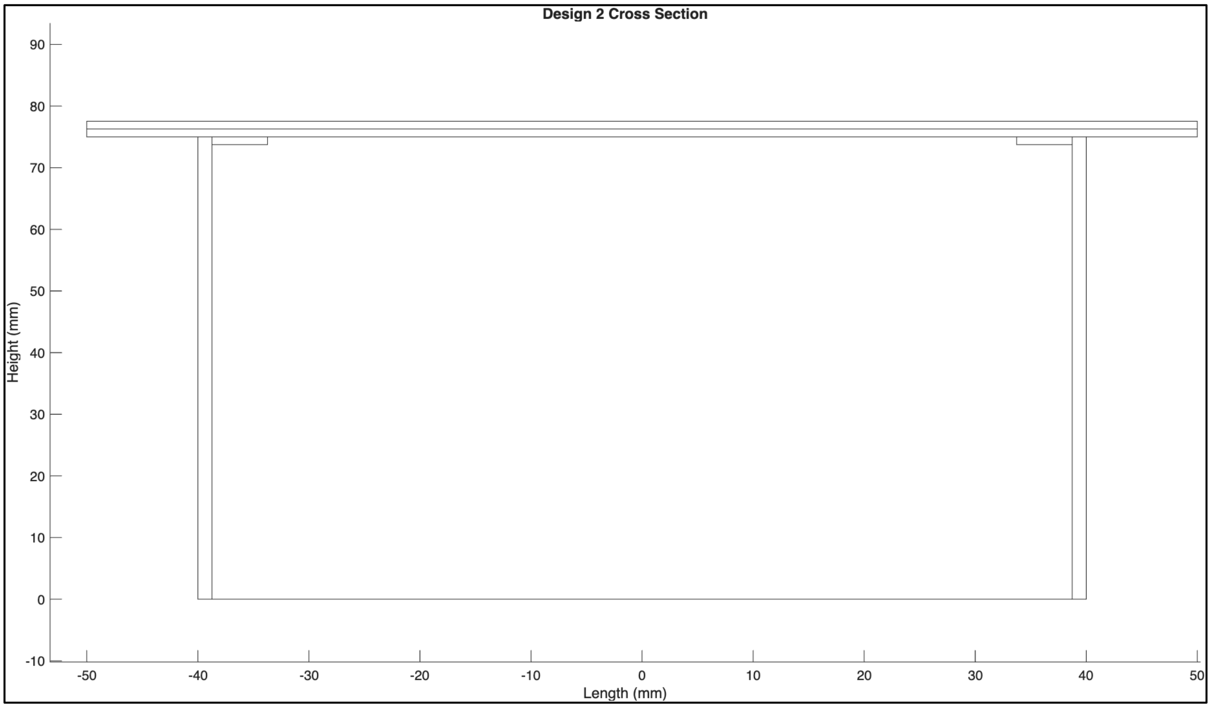

Figure 5: Image displaying the cross-section of iteration 2.

Important Metrics

Table 5: Table showing the important metrics of iteration 2

Maximum Load (N)	506
Percent of Material Used	59.21
Efficiency Index	0.0816
Centroidal Axis From Bottom (mm)	60.1
Moment of inertia ($\times 10^6 mm^4$)	0.256

Table 6: Table showing all factors of safety of iteration 2 with the most likely failure mode highlighted

Compressive Failure of Top Flange	1.266
Tensile Failure of Top Flange	1.842
Shear Failure at Centroidal Axis	2.21
Shear Failure of Glue	4.83
Plate Buckling Case 1	2.92
Plate Buckling Case 2	19.83
Plate Buckling Case 3	36.7
Plate Buckling Case 4	2.81

The entirety of the bottom piece was removed, reducing material usage, improving the strength to weight ratio, without significantly hindering structural performance. This change shifted the $\bar{y}$ upwards to 60.1mm and reduced I to $2.56 \times 10^5 mm^4$. This reduced the factor of safety for the compressive failure of the top flange from 1.674 to 1.266 since the second moment of inertia was reduced and the flexural compression is given by ($\sigma = \frac{My}{I}$). Additionally, this reduced the factor of safety of case 1 plate buckling to 2.92 from 3.85, as the flexural stress ($\sigma = \frac{My}{I}$) the top flange experiences increased. This means that flexural compression in the top flange is still the limiting failure mode, and the the maximum load capacity is now reduced to 506N and the efficiency index has decreased slightly to 0.0816. The factor of safety of the tensile strength of the walls decreased to 1.842 from 4.73, this was because removing the base significantly reduced the I, meaning the applied stress on the bottom increased, reducing the factor of safety. The factor of safety of case 3 plate buckling almost doubled, mostly due to the increase in $\bar{y}$, decreasing the amount of the web in compression, meaning the maximum applied compressive stress also decreased dramatically, increasing the factor of safety.

Iteration 3: Increase the Height

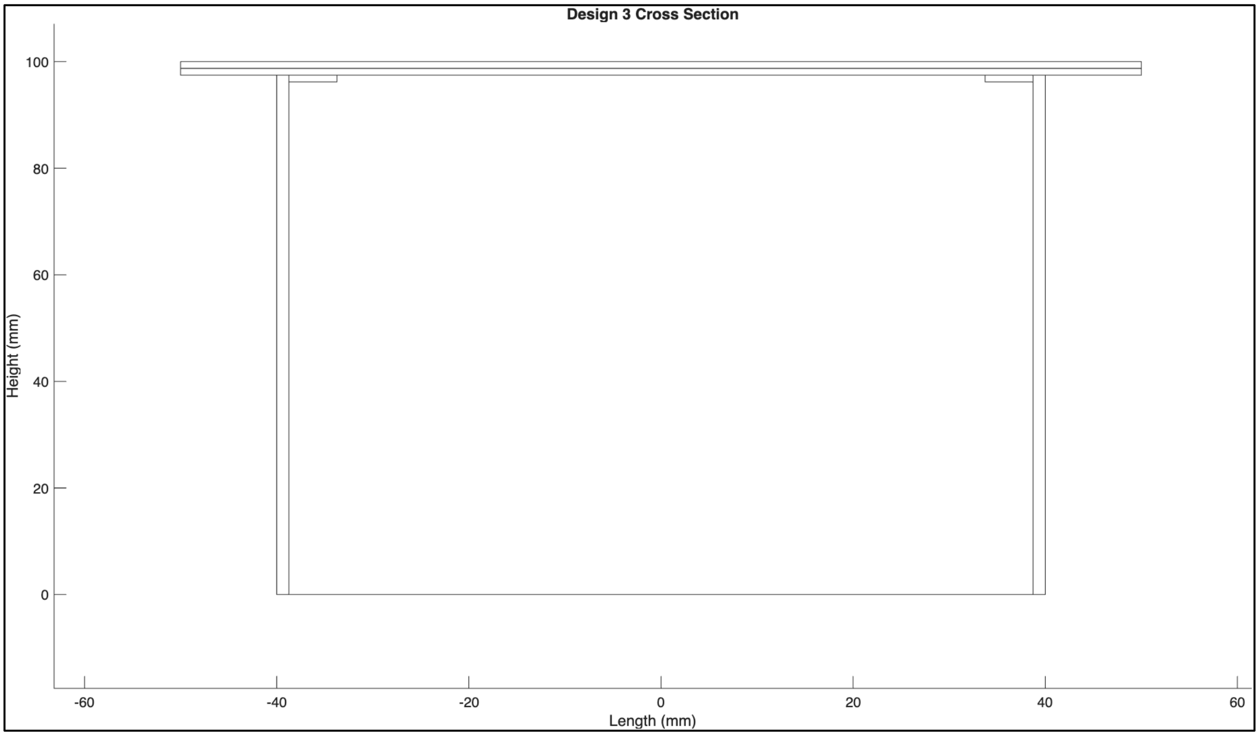

Figure 6: Image displaying the cross-section of iteration 3.

Table 7: Table showing the important metrics of iteration 3 with the failure mode highlighted.

Maximum Load (N)	703
Percent of Material Used	67.2
Efficiency Index	0.0997
Centroidal Axis From Bottom (mm)	74.6
Moment of inertia ($\times 10^6 mm^4$)	0.516

Table 8: Table showing all factors of safety of iteration 3 with the most likely failure mode highlighted.

Compressive Failure of Top Flange	1.76
Tensile Failure of Top Flange	2.99
Shear Failure at Centroidal Axis	2.88
Shear Failure of Glue	6.55
Plate Buckling Case 1	4.05
Plate Buckling Case 2	27.5
Plate Buckling Case 3	20.7
Plate Buckling Case 4	2.22

The total height of the cross-section increased from 77.54mm to 100mm, addressing the compression on the top flange. As I is directly proportional to the cube of height ($I \propto h^3$), increasing the height raised I to 5.16$\times 10^5 mm^4$, lowering the applied compression on the top flange, increasing the factor of safety of compressive failure to 1.760 from 1.266. The maximum load therefore increased to 703N and the efficiency index increased to 0.0997, however compressive failure on the top flange is still the limiting failure mode. The factor of safety of tensile failure also increased, to 2.99 from 1.842, due to the increased I, significantly decreasing the applied tensile stress on the walls, increasing the factor of safety. The factor of safety of case 3 plate buckling decreased to 20.7 from 36.7, mainly due to the increase in height, as the buckling stress is directly proportional to the square of height ($\sigma \propto h^2$), meaning an increase in height will lower the factor of safety, however as the factor of safety was previously the highest one, this is not a problem.

Iteration 4: Lower length between webs (lower base width)

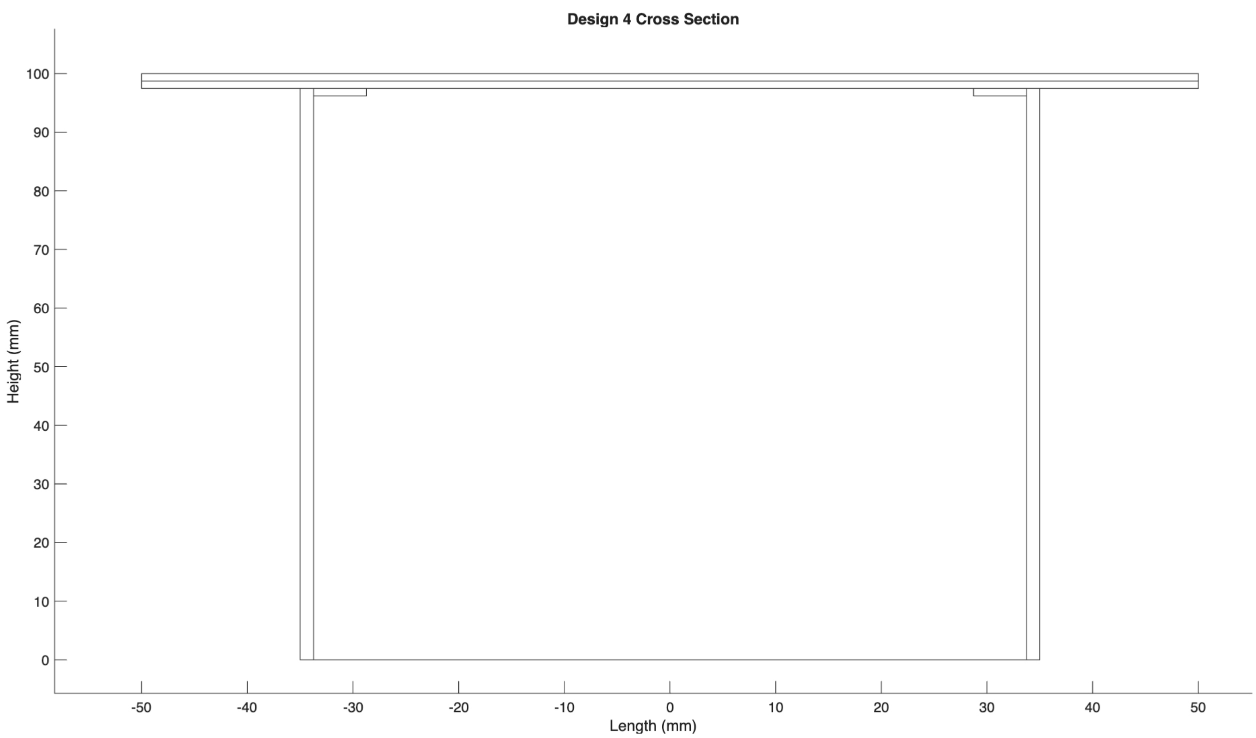

Figure 7: Image displaying the cross-section of iteration 4.

Table 9: Table showing the important metrics of iteration 4.

Maximum Load (N)	703
Percent of Material Used	66.5
Efficiency Index	0.1008
Centroidal Axis From Bottom (mm)	74.6
Moment of inertia ($\times 10^6 mm^4$)	0.516

Table 10: Table showing all factors of safety of iteration 4 with the most likely failure mode highlighted.

Compressive Failure of Top Flange	1.76
Tensile Failure of Top Flange	2.99
Shear Failure at Centroidal Axis	2.88
Shear Failure of Glue	6.55
Plate Buckling Case 1	5.29
Plate Buckling Case 2	12.24
Plate Buckling Case 3	20.7
Plate Buckling Case 4	2.22

The base width of the design was shortened from 80mm to 70mm by lowering the distance between the webs, this was done to increase the factor of safety of case 1 plate buckling, decrease the extremely high factor of safety for case 2 buckling, and decrease the area used up by the diaphragms. This change lowered the width *b* of the top plate, increasing the flexural stress capacity according to the equation $\sigma_{buck} = \frac{4\pi^2 E}{12(1-\mu^2)} \times (\frac{t}{b})^2$. The $\bar{y}$ and I did not change, however the factor of safety of case 1 plate buckling increased from 4.05 to 5.29. Although the maximum load did not change because compressive failure of the top flange was still the limiting factor, the efficiency increased to 0.1008 from 0.0997. The factor of safety of case 2 plate buckling decreased to 12.24 from 27.5, as the overhang length b increased to 15mm each, and as buckling stress is inversely proportional to b, the applied buckling stress increased, reducing the factor of safety. All the other factors of safety remained the same, as there was no change in the value of I or $\bar{y}$.

Iteration 5: Increase height (again)

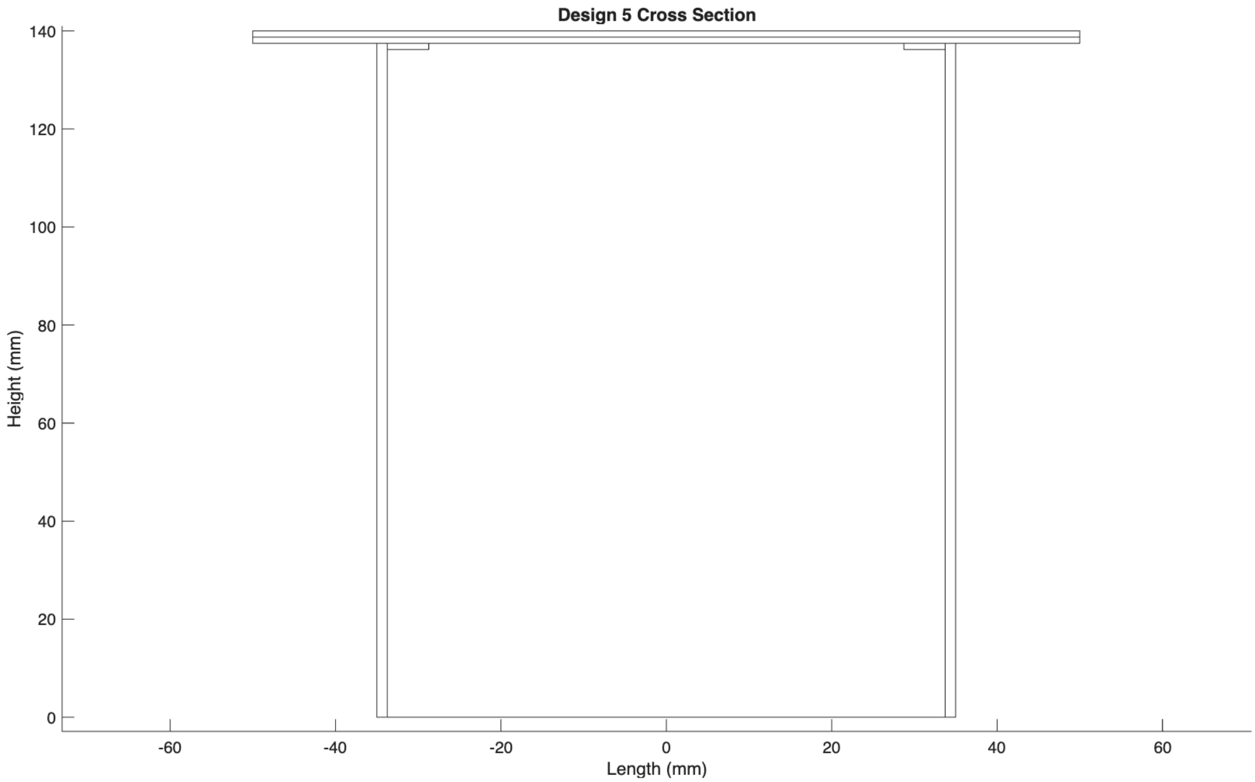

Figure 8: Image displaying the cross-section of iteration 5.

Table 11: Table showing the important metrics of iteration 5.

Maximum Load (N)	669
Percent of Material Used	80.7
Efficiency Index	0.0791
Centroidal Axis From Bottom (mm)	99.0
Moment of inertia ($\times 10^6 mm^4$)	1.289

Table 12: Table showing all factors of safety of iteration 5 with the most likely failure mode highlighted.

Compressive Failure of Top Flange	2.72
Tensile Failure of Top Flange	5.63
Shear Failure at Centroidal Axis	4.09
Shear Failure of Glue	9.94
Plate Buckling Case 1	8.18
Plate Buckling Case 2	18.93
Plate Buckling Case 3	10.84
Plate Buckling Case 4	1.674

The total height was increased again, from 100mm to 140mm, again to address case 1 plate buckling. This increased I to $1.289 \times 10^6 mm^4$, drastically lowering the flexural stress across all members, improving the factor of safety of case 1 plate buckling to 8.18 from 5.29 as $FOS = \frac{\sigma_{capacity}}{\sigma_{applied}}$. The factor of safety of tensile stress increased to 5.63 from 2.99, this is mainly as the I increased significantly, reducing the applied tensile stress applied to the bottom of the cross-section, increasing the factor of safety. The factor of safety of case 2 plate buckling increased dramatically, from 12.24 to 18.93, this was due to the increase in I reducing the applied buckling stress as they are inversely proportional, this increase in applied buckling stress increased the factor of safety. The factor of safety of case 4 plate buckling stress decreased from 2.22 to 1.674, becoming the limiting failure mode of the design, this was as the buckling stress is inversely proportional to the height of the webs according to the equation $\sigma_{buck} = \frac{5\pi^2 E}{12(1-\mu^2)} \times \left(\left(\frac{t}{h}\right)^2 + \left(\frac{t}{a}\right)^2\right)$, meaning the increase in height causes a decrease in the buckling stress, causing the decrease in the factor of safety. As such, the maximum load and efficiency index fell to 669N and 0.0791 respectively.

Iteration 6: Decrease Diaphragm Spacing and Increase # of Diaphragms

```
%%% Define the Properties of the CrossSection %%%
thickness = 1.27;
basewidth = 70; %%% Must be Able to Hold The Trains
basethickness = 0;
topflangethickness1 = thickness;
topflangethickness2 = thickness;
totalbeamheight = 140;
topwidth = 100;
topwidthsecondlayer = 100;
tabwidth = 5;
%%% ----------------------------------------- %%%
%%%DEFINE THE DIAPHRAGMS
DiaphragmSpacing = 150;  %%% Max Diaphragm Spacing
NumberDiaphragms = 11;
%%% ----------------------------------------- %%%
```

Figure 9: An image showing the parameters used when running the code to evaluate the final design.

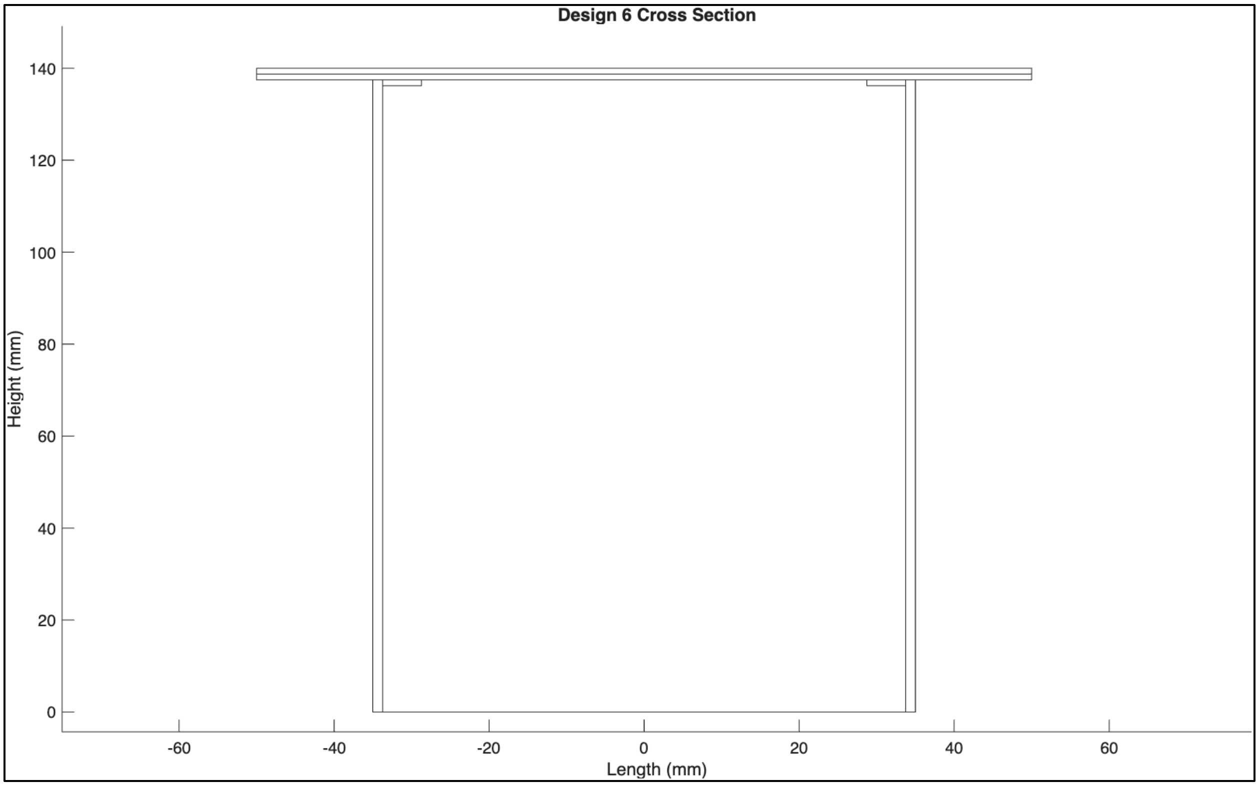

Figure 10: Image displaying the cross-section of iteration 6.

Table 13: Table showing the important metrics of iteration 6 (Load Case 1).

Maximum Load (N)	1088
Percent of Material Used	86.3
Efficiency Index	0.1202
Centroidal Axis From Bottom (mm)	99.0
Moment of inertia ($\times 10^6 mm^4$)	1.289

Table 14: Table showing all factors of safety of iteration 6 (Load Case 1) with the most likely failure mode highlighted.

Compressive Failure of Top Flange	2.72
Tensile Failure of Top Flange	5.63
Shear Failure at Centroidal Axis	4.09
Shear Failure of Glue	9.94
Plate Buckling Case 1	8.18
Plate Buckling Case 2	18.93
Plate Buckling Case 3	10.84
Plate Buckling Case 4	2.75

Load Case 1

The distance between diaphragms was reduced from 400mm to 150mm, increasing the total number of diaphragms to 11, this was done to address case 4 plate buckling, which was the limiting failure mode of iteration 5. Due to this change the factor of safety of case 4 plate buckling increased to 2.75 from 1.670, increasing our maximum load to 1088N, increasing the efficiency index to 0.1202. This increase was caused by the decrease in distance between the diaphragms (a), the square of this distance is inversely proportional to buckling stress ($\sigma_{buck} \propto \frac{1}{a}$), therefore reducing a caused the buckling stress capacity to increase, increasing the factor of safety. As this change did not affect I or $\bar{y}$, none of the other factors of safety changed at all, meaning that compressive failure of the top flange is the limiting failure mode, however as this design has a very high predicted strength to weight ratio, and a maximum load capacity over 1kN, this was the design chosen to be built, and therefore the final iteration.

Load Case 2

Table 15: Table showing the important metrics of iteration 6 (load case 2).

Maximum Load (N)	1053
Percent of Material Used	86.3
Efficiency Index	0.1163
Centroidal Axis From Bottom (mm)	99.0
Moment of inertia ($\times 10^6 mm^4$)	1.289

Table 16: Table showing all factors of safety of iteration 6 with the most likely failure mode highlighted.

Compressive Failure of Top Flange	2.42
Tensile Failure of Top Flange	5.00
Shear Failure at Centroidal Axis	3.46
Shear Failure of Glue	8.41
Plate Buckling Case 1	7.28
Plate Buckling Case 2	16.84
Plate Buckling Case 3	9.64
Plate Buckling Case 4	2.33

The final design is now analysed using load case 2 instead of load case 1. I and $\bar{y}$ remain unchanged, as no dimensions were changed. Overall, all the factors of safeties decreased due to the increased load in load case 2, with case 4 plate buckling now being the limiting failure mode, with a factor of safety of 2.33, due to the increased maximum shear force caused by load case 2 having unevenly distributed weight across the train cars. The maximum load capacity therefore reduces to 1053N due to this difference in the load distribution, that results in a greater maximum shear and a greater maximum moment.

Bibliography

[1] University of Toronto, “CIV102 Matboard Bridge Design Project – Fall 2025,” Table 2: Specified Material Properties, pp. 8

[2] L. EP Latex / Lepage, “Contact Cement Technical Data Sheet,” Google Drive